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Boundary Spanning Roles and Organization Structure¹

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Boundaries are a defining characteristic of organizations, and boundary roles are the link between the environment and the organization. The creation, elaboration, and functions of boundary spanning roles are examined, with attention to environmental and technological sources of variation in the structure of boundary roles. Eleven hypotheses integrate the material reviewed and are amenable to empirical test. Future research should overcome problems created when organizations are treated as "wholes" or single entities.

A minimal defining characteristic of a formal organization is the distinction between members and non-members, with an organization existing to the extent that some persons are admitted,

while others are excluded, thus allowing an observer to draw a boundary around the organization (61, pp. 139-146). Defining organizations in

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This review could not have been written without the foundation laid by James D. Thompson, and our debt to him should be evident to those familiar with his work.

terms of boundaries to interaction also allows a parsimonious definition of the role of formal authority in an organization: authorities are persons who apply organizational rules in making decisions about entry and expulsion of members (6, p. 283). In this sense, organizational behavior (OB) has always contained an implicit "open systems" view, although few theorists or researchers have studied boundary-maintaining or boundary-crossing (5).

The definition and location of a specific boundary may be possible only given a specific conceptual and empirical context. This article takes the existence of boundaries as given, while treating boundary spanning activity as problematic. Specifically, it examines functions served by boundary roles, the generation of boundary units and roles relating organizations to their environments, and the environmental and organizational sources of variation in the structure of boundary roles. Use is made of existing literature, but the argument is speculative at many points.

Although most investigators agree on the importance of focusing on relations between organizations and their environments, there is little agreement on the degree of autonomy of action organizations have vis-à-vis their environments. At the extreme are two positions: a natural selection model, laying heavy emphasis on the dominance of environmental constraints on behavior, and a strategic choice or resource dependence model, emphasizing the active role organizational administrators play in shaping outcomes. As these two macro-theoretical positions are reviewed elsewhere (9), and there is no prospect of reconciling them in this paper, we concentrate on propositions and hypotheses of the middle-range.

Functions of Boundary Roles

Two classes of functions are performed by boundary roles: information processing and external representation. Information from external sources comes into an organization through

boundary roles, and boundary roles link organizational structure to environmental elements, whether by buffering, moderating, or influencing the environment. Any given role can serve either or both functions.

The Information Processing Function

In focusing on the information processing function, we are following the lead of Dill (24), who suggested that the environment of an organization could be treated as information available to the organization through search or exposure. Thompson and McEwen discussed the organization's need for information to judge the amount and sources of support for its goals (63, p. 30). Terryberry argued that viable organizations are characterized by "an increase in the ability to learn and to perform according to changing contingencies in the environment" (60, p. 660).

Boundary role incumbents, by virtue of their position, are exposed to large amounts of potentially relevant information. The situation would be overburdening if all information originating in the environment required immediate attention. Boundary roles are a main line of organizational defense against information overload (17, 42). Expertise in selecting information is consequential, since not all information from the environment is of equal importance. External information can be conceptualized in terms of a three-part hierarchy, corresponding to Parsons' (47) distinction between three levels of authority in organizations: strategic, managerial, and technical information (17, p. 325). Their relative importance varies by type of environment and technology; e.g., in stable homogeneous environments and organizations with highly routinized technology, strategic information is less important.

The process by which information filters through boundary positions into the organization must be examined. Boundary roles serve a dual function in information transmittal, acting as both filters and facilitators. Information over-

load would still be a problem if all relevant information had to be immediately communicated to internal members. Accordingly, boundary role personnel selectively act on relevant information, filtering information prior to communicating it (23, 40). They act autonomously on some information, and consolidate, delay, or store other information, thus alleviating the problem of overloading communication channels (although perhaps incurring other costs to the organization in the process). Information is summarized and directed to the organizational units that need it.

Boundary role personnel may act on information requiring an immediate response, as when a sales department responds to a customer inquiry about product specifications. They may store information for future use, as when a purchasing department files information on a new supplier's products, to be referred to at re-order points. Boundary personnel in marketing may uncover trends in the demand for their organization's products which will have a major impact on the mix of resources required, and communicate this information to purchasing. Boundary units may also summarize information and communicate it to other units on a regularized basis (41).

The expertise of boundary role occupants in summarizing and interpreting information may be as important to organizational success as expertise in determining who gets what information, depending upon the uncertainty in the information processed. Information to be communicated often does not consist of simple verifiable "facts". If the conditions beyond the boundary are complexly interrelated and cannot be easily quantified, the boundary role incumbents may engage in "uncertainty absorption", — drawing inferences from perceived facts and passing on only the inferences (40, p. 189).

Consider the case of a lobbyist formulating a report on a bill and amendments that will differentially affect the operations of his or her organization. The lobbyist will have to summarize information about the bill's progress, testimony in

hearings, and apparent predispositions of committee members and other legislators, as well as making the entire situation meaningful to his or her superiors. If these superiors cannot understand the interrelationships and implications of the raw data, they will not be able to use the information. Some simplification is necessary and the relationships of events in Washington to organizational operations will have to be clearly specified. In short, the lobbyist must put the information in usable form (67).

Innovation and structural change are often alleged to result from information brought into the organization by boundary personnel (11, 28). All complex organizations have a tendency to move toward an internal state of compatibility and compromise between units and individuals within the organization, with a resultant isolation from important external influences (18). This trend can jeopardize the effectiveness and perhaps the survival of the organization, *unless* the organization is effectively linked to the environment through active boundary personnel. By scanning the environment for new technological developments, innovations in organizational design, relevant trends in related fields, etc., boundary personnel can prevent organizations from becoming prematurely ossified and mismatched with their environments (20).

This review of the information processing function of boundary roles may be summarized in the following hypothesis:

- H1: *An organization's ability to adapt to environmental contingencies depends in part on the expertise of boundary role incumbents in selecting, transmitting, and interpreting information originating in the environment.*

The External Representation Function

External representation can be viewed in terms of an organization's response to environmental influence. Environmental constraints and contingencies can be adapted to in at least three ways: (a) by internal structural differentiation to

match the pattern of the relevant environment, which requires information about environmental characteristics; (b) by gaining power over relevant elements of the environment, manipulating it to conform to the organization's needs; and (c) a compromise position, the modal pattern of use of boundary personnel in "normal" boundary spanning roles. Included under the external representation function are all boundary roles that involve resource acquisition and disposal, political legitimacy and hegemony, and a residual category of social legitimacy and organizational image.

Boundary roles concerned with resource acquisition and disposal include purchasing agents and buyers, marketing and sales representatives, personnel recruiters, admissions officers, and shipping and receiving agents. In these roles the organization is represented to the environment, because the normal flow of authoritative commands is from the core of the organization to these boundary roles. The behavior of personnel in these roles is supposed to reflect the policy decisions of decision makers in line roles.

This usual flow of directives to boundary roles presents two problems for boundary personnel. First, much of the information they attend to has an external origin, and it occasionally becomes apparent that policy directives are based on information that is no longer relevant. This poses a dilemma for the conscientious boundary spanner, especially in organizations with a high degree of decentralization — should behavior be immediately modified to correspond to latest developments, or should action be delayed until the information has "gone through channels?" Second, as Strauss (59) pointed out in his study of purchasing agents, some boundary personnel are not satisfied with their subordinate position on the vertical axis of the organization, given their self-evident horizontal location of equality with other departments. Thus dissatisfied boundary spanners take the initiative to increase their power vis-à-vis other units. For example, personnel officers suggest changes in job descriptions before agreeing

to post them; admissions or intake staff develop their own criteria of "worthy" applicants (14); and purchasing agents make informal compacts with salespersons from outside firms to push products which their production department "really" needs (59).

Boundary roles involved with maintaining or improving the political legitimacy or hegemony of the organization not only represent the organization but also mediate between it and important outside organizations. The term "mediate" refers here to aspects of the boundary role involving negotiations that will eventually affect the power of the focal organization vis-à-vis another organization or group. Kochan (36) notes that city governments have created collective bargaining units as a response to threats to the city's control over its employees. The role of the corporate lawyer is perhaps the most clear example of the necessity and difficulty of preserving at least an equal balance of rights and responsibilities between business organizations (38).

Boundary spanning personnel can help maintain the legitimacy of the organization by providing information to important client groups, specially adapted for them. Aldrich and Reiss (10) note that police officers on the beat transmit an image of city law enforcement capabilities to small *businesspersons* independent of the *businessperson's* attitudes toward the police themselves. Information transmittal is facilitated because both police and small business are exposed to environmental forces that make their commonality of interest highly salient. Adair's (1) study of the use of Navaho Indians as health aides for their own communities found that the Indians functioned as mediators in their boundary roles, drawing the doctor and the Navaho patient closer together. The Indian health worker offers a different side to each party involved, finding a middle ground to settle discords between them. Detached school workers perform the same sort of representation function for school systems.

Maintaining the organizational image and

enhancing its social legitimacy are less a matter of mediating contacts than of simply making the organization visible. Advertising and public relations specialists try to influence the behavior of target groups in ways that benefit the focal organization, without bargaining or negotiating with the target group. The flow of intra-organizational influence to these roles is much more one-sided than in the boundary roles described above; one apparent consequence is a high rate of turnover.

One function of boards of directors and public advisory commissions is to link the organization to target groups in the environment in a highly visible way, so that they will feel their interests are being represented. Thus, women and blacks are being appointed in increasing numbers to corporate boards, and students now serve as trustees on university boards. Fulfilling this function requires recruiting people who are already members of or in contact with specific target groups. Maniha and Perrow (39) describe the formation of a Youth Commission's board in terms of such community interest groups as Catholics, the university, and youth-oriented voluntary associations.

These three varieties of external representation functions can be related to organizational effectiveness in the second hypothesis:

- H2: *An organization's ability to cope with environmental constraints depends in part on the ability of boundary role incumbents to achieve a compromise between organizational policy and environmental constraints, to choose strategic moves to overcome constraints, or to create conditions in which the organization's autonomy is seldom challenged.*

Creation of Boundary Roles

By definition all organizations have some boundary spanning roles, if only at the level of the organization head or chief executive. But some have an elaborate set of boundary roles

while others have only a few. In some cases boundary roles are formalized into full-time organizational positions, while in others they are only part-time activities. This section examines the generation and formalization of boundary roles as explicit organizational roles, with references to organizational size and technology, and various characteristics of organizational environments. To understand the process of boundary spanning *behavior*, an interactive model of the kind developed by Adams (2) is needed, but such models are highly specific to the particular pair-wise relationship being examined. Here we are concerned with the general features of the boundary role while recognizing that actual behavior in boundary roles will vary from context to context.

The extent to which organizational positions involve interaction with external elements varies greatly. Many positions outside the technical core involve some extra-organizational interaction, but only a few involve intensive interaction. As an empirical test of arguments in this article we would need a measure providing better than a "yes-no" categorization of roles as either boundary spanning or not. This would require determining the proportion of time spent with outsiders, the number of outside contacts, the importance of each contact, etc., as Whetten (65) has recently done in a study of manpower organizations.

The number of formally designated boundary spanning roles in an organization is partially dependent on organizational size (5). A small organization is able to survive with a fairly simple structure, with relatively few differentiated roles and functions (16, 21, 22). Its structure may be less formalized and more amenable to restructuring to achieve and maintain a satisfactory position vis-à-vis its environment. A small organization might be willing and able to rely on information brought to it informally by its members. This tendency is more marked among organizations that have highly committed members or that are not highly dependent on their environment for survival (6), such as a small religious sect

(29). As organizational and environmental complexities increase, organizations can no longer afford non-differentiated boundary spanning activities.

Technology and Boundary Role Differentiation

Holding size constant, boundary spanning units or roles should be expected to increase as a proportion of all roles as organizations differentiate in response to the interaction of technology and environment, and under the direct impact of environmental pressure. In the following discussion, technology is treated as a source of internal differentiation generating boundary roles to the extent that varying technology types create different patterns of organization-environment interaction. Thompson's categories of mediating, long-linked and intensive technology capture the implications of various technology types for the generation of boundary roles (62, pp. 15-18).

Organizations with a *mediating technology* link clients or customers with each other, as in the case of banks, insurance companies, or the post office; or they link clients with other organizations, thus serving a "people-processing function" (30). Such organizations should have the highest proportion of boundary roles, as boundary roles are their line roles. The wholesaler of small consumer goods has boundary personnel who purchase goods from producers and sell them to organizations which, in turn, sell them to retail customers. An investment banking firm contracts with a client to put together a "package" of investment instruments that satisfy financial needs and then sells the "package" to other organizations with funds to invest. Boundary personnel similarly serve a line function in the people processing component of organizations such as schools and government agencies.

A study of organizations using a mediating technology in the book publishing and record producing industries found that they allocated a large proportion of their personnel to boundary

spanning roles, on the input side to contract for and supervise the production of raw material, and on the output side to promote the cultural products and achieve optimal distribution (32). These boundary roles also monitored the environment and provided information quickly to managers and executives, apparently as a strategy to help the organization hold its position in a very uncertain environment.

Organizations with a *long-linked technology* attempt to buffer most of their units and roles from the environment, and have a lower proportion of boundary roles. Since the various organizational units are serially interdependent, there are many boundary roles between intra-organizational components, but the focus here is on roles at the external boundary. Specific boundary roles are important for such organizations. First, long-linked technology gains maximum efficiency through standardized production of large volumes of output (to take advantage of economies of scale) and so such organizations need an effective marketing and sales force (35, 51). Second, Thompson (62, p. 40) argues that "organizations employing long-linked technologies and subject to rationality norms seek to expand their domains through *vertical integration*", and thus the legal and accounting departments of such organizations interact fairly intensively with potential acquisitions in the environment (50).

Organizations using an *intensive technology*, which depends on the object being worked on, also buffer most of their roles from the environment. They often achieve this by temporarily drawing the object or the client into the organization. In intensive technology organizations concerned with people-changing activities, the client is temporarily assigned an organizational role, and must change behavior to suit norms which preclude appealing to his or her environmental role relative to the organization. The boundary personnel who engage in initial interaction with potential clients affect the organization's subsequent internal operations if they have the power to admit or reject clients, e.g.,

the physician associated with a hospital or the admissions officer at a private college. By detecting a violation and making an arrest, a police officer provides the rest of the criminal justice system with raw material to be processed (55).

Thompson argues that these organizations seek to expand their domains by incorporating the object worked on, with "total institutions" (26) placing an almost impenetrable boundary around clients. The people-changing organizations that use an intensive technology (e.g., hospitals) have one characteristic that opens them to environmental influence — their high degree of professionalization. Aiken and Hage (4) assert that professionals in organizations engage in a great deal of boundary-spanning contact because of the need to maintain contact with a professional reference group and keep abreast of changing technology in their field.

In intensive technology organizations not concerned with people-changing activities, the clients often become a temporary part of the organization's administrative structure and thus need a liaison person to represent them, as in the construction industry (62, p. 44). Thus, while it is clear that mediating technology organizations have proportionately more boundary spanning roles than other organizations, the relative ranking of organizations with long-linked and intensive technology cannot be determined without further empirical research.

H3: *Organizations using a mediating technology will have the highest proportion of boundary roles, while organizations using long-linked and intensive technologies will have a smaller proportion of boundary roles.*

H4: *Organizations using long-linked or intensive technologies will departmentalize and otherwise separate boundary spanning units from their core technical units.*

Environment and Boundary Role Differentiation

Environmental pressures are responsible for much of the observed differentiation in organi-

zations, after technology is taken into account. Some theories of organization-environment interaction posit that maintenance of a high degree of internal organizational complexity occurs only in response to environmental pressures that tolerate nothing less (9, 18). The concentration of important environmental elements into an organized form may promote a matching organizational response, in the form of more boundary units or more formalized and centralized boundary spanning activities (52). Kochan points out that:

A number of collective bargaining researchers have noted the proliferation of specialized labor relations units in city governments in response to the increased unionization of public employees (36, p. 7).

Wilson (67) discusses the growth of lobbying efforts of unions, trade associations, and other organizations representing vested interests, in response to the growth in power of the federal government. The consumer, ecology, and other movements have brought pressure on corporations, which have responded by establishing public relations units to deal with pressure groups. The same type of response occurred among public agencies in the President's Office for Consumer Affairs and similar offices in HEW, HUD, etc. (45).

H5: *Organizations in environments where important elements are concentrated will have a higher proportion of boundary roles than organizations in environments where important elements are dispersed.*

Heterogeneous environments should evoke more organizational boundary spanning units and roles, as organizations "seek to identify homogeneous segments and establish structural units to deal with each" (62, p. 70). Separate units, whether established on the basis of heterogeneity in a client population or in the geographical domain served, lead to a higher proportion of boundary roles than in organizations of comparable size serving a homogeneous domain. Hospitals establish separate units for obstetrics, con-

tagious diseases, and out-patient services. Auto manufacturing firms respond to heterogeneity in client income distribution by divisionalizing operations around products with different selling prices, but not necessarily costs (19).

H6: *Organizations in heterogeneous environments will have a higher proportion of boundary roles than organizations in homogeneous environments.*

Stable environments, which presumably call for less frequent monitoring, should evoke fewer boundary roles than unstable environments, although much depends on whether change is occurring at a constant or variable rate (62, pp. 71-72). In the cultural industry (books, records, films) where fashions change rapidly, we would expect to find proliferation of boundary roles on both input and output sides of the organization (32). In organizations producing for a stable market, we would expect most roles to be related directly to the production process, although an unexpected shift in the market can change the situation dramatically, as in Emery and Trist's (25) example of a food canning firm. Some theorists have argued that the most salient characteristic of organizational environments today is their rate of change (60), a purported trend which should cause an increase in the proportion of boundary spanning roles in most organizations.

H7: *Organizations in rapidly changing environments will have a higher proportion of boundary roles than organizations in stable environments.*

A final dimension to be considered is the extent to which the environment is rich or lean in resources (7, 12, 25). In rich environments, holding competition constant, we would expect to find fewer boundary roles, since environmental search and monitoring would be less critical for organizational survival than in environments where lack of resources prevents the accumulation of a "resource cushion". At a time of international or interorganizational hostility, the environment becomes less rich in information and

so nations and organizations have to allocate more roles to their boundaries to make use of what little information is available (66).

H8: *Organizations in lean environments will have a higher proportion of boundary roles than organizations in rich environments.*

Environment and Boundary Role Formalization

If internal complexity is associated with environmental pressures and demands, organizational boundary roles will be officially recognized full-time roles, especially if decision makers recognize the existence of such contingencies. Whether boundary roles will be thus formalized depends upon organizational recognition of potentially costly contingencies that may arise from failure to maintain effective links to elements in the environment. But such recognition need not be based on intelligence that organization itself has accumulated, as professional education (e.g., MBA programs), professional and trade publications (e.g., *Business Week*), and informal inter-firm contact (51) all keep organizational decision-makers abreast of new developments in the design and administration of formal organizations (12, 58). The following discussion focuses on direct, rather than indirect, recognition of environmental contingencies and constraints.

Most large organizations formally designate such roles as labor negotiators and corporate lawyers responsible for transactions in the labor relations sector, since strikes and law suits might cripple an organization. Labor contracts are negotiated for fairly long periods of time and the organizational costs of mistakes in boundary-spanning negotiation with unions are fairly high. Smigel's (57) discussion of staff recruitment in large Wall Street law firms indicates that firms became aware of a variety of changes in their environment, including the small output of prestigious law schools, students' wariness of accepting positions with large firms, and the increasing demand for trained lawyers. One result was crea-

tion of the formal position of "hiring partner" to scan the potential output of top ranking law schools, sell students on advantages of employment in the firm, and thus improve recruitment of desirable graduates.

The more critical the contingency, the more attention is paid to explicit formalization of the role and selection of an incumbent. This is particularly evident with regard to the composition of boards of directors of large organizations, as Pfeffer argues that organizations:

use their boards of directors as vehicles through which they co-opt, or partially absorb, important external organizations with which they are interdependent (50, p. 222).

Price (53), in a study of state wildlife governing boards, found that one major function of board members was to serve as a buffer group between the full-time staff and the public. Zald points to the external representation functions of boards of directors:

They promote and represent the organization to major elements of the organizational set, for example, customers, suppliers, stockholders, interested agencies of the state, and the like. That is, they defend and support the growth, autonomy, and effectiveness of their agencies vis-à-vis the outside world (68, p. 99).

Another critical contingency for large corporations involves managing reciprocal relations with other large firms; a trade relations person is alerted to look for opportunities to cooperate with other firms when it could be to their mutual advantage. Perrow notes that:

the practice of reciprocity is so extensive that about 60 percent of the top 500 corporations have staff members who are explicitly assigned to trade relations. Of course, any smart sales executive or top executive can serve in this capacity. However, it is striking that the practice is sufficiently common to justify so many special positions among the giants (48, p. 122).

Pursuing leads on possible acquisition of other companies is an important function assigned to corporate development units. Aguiler notes that the high volume of acquisition leads generated by this staff:

demonstrates how the formalization of a search procedure can significantly increase a company's relative involvement with a particular kind of information (3, p. 47).

- H9: *Boundary roles are most likely to be formalized when crucial environmental contingencies have been explicitly recognized by organizational decision makers, or the organization is structured in a way that facilitates the adoption of structural innovations through imitation and borrowing from other organizations or other external sources.*

Routinization, Discretion, and Power

The degree of role specificity (31) of boundary roles varies considerably, with some being highly routinized and others highly non-routine. Thompson (61) identified two conditions leading organizations to increase specificity of control over boundary role personnel. First, organizations that provide services for large numbers of persons and thus face many non-members (relative to members) at the boundaries of the organization must either substantially increase the number of personnel in boundary positions or else routinize the tasks of existing staff so they can handle a higher volume of work. Second, organizations using a mechanized production technology which places a premium on large runs of standardized products depend upon a large volume of standardized transactions per member at the organization's output boundary. Pressures for routinization are somewhat lessened when the non-members dealt with have little or no discretion to participate in a relationship. Later Thompson identified a third condition, in that stable environments are likely to produce boundary roles governed by rules, whereas unstable environments are likely to increase flexibility in the specificity of boundary role routines (62, p. 71).

Purchasing agents and sales personnel interact frequently with suppliers and buyers and usually deal with fairly homogeneous groups of

organizations and individuals. A high frequency of interaction and homogeneity of elements at the boundary allows behavior in these roles to be highly routinized (27, p. 24; 62, p. 111). Routinization is reflected in the existence of standard purchase and sales forms or contracts, standard operating procedures for soliciting and accepting bids, and standard operating procedures for calling on customers and closing sales. A classic example is the retail salesperson who knows the one proper way to record a cash sale and the one proper way to record a credit transaction.

Routinization of roles at the organization's boundary not only increases efficiency in handling predictable relationships and large numbers of repetitive transactions, but also serves a social control function. The programmed nature of these activities is partial insurance of boundary spanner consistency with organizational procedures, norms, and goals. Members who interact freely with non-member groups, particularly homogeneous sets, are likely to develop attitudes consistent with those of the non-members, rather than of their focal organization. The existence of standard operating procedures partially protects the organization against attitudes and behaviors that are not consistent with organizational objectives.

MacAuley (38) noted different behavioral orientations among boundary and non-boundary personnel in the use of contracts among business firms. Sales departments tended to display non-organizational norms that made them willing to conduct transactions without legally binding contracts. Members of the controllers and legal departments upheld the organizational norm of using contracts, which also happened to be the norm of their professional reference group.

Mathiesen's (41) study of prison staff members identified the boundary role of "social worker" as a position that was difficult to routinize.

Though almost all staff members claimed there were few or no specific rules or regulations guiding their communications, the social

workers appeared particularly vehement about it, and included relations to official organizations. They stressed that here they had to be extremely flexible; that they had to organize the work on a day-to-day basis and according to the unique circumstances of the individual case (41, p. 76).

Telephoning was preferred to the use of letters, and when complex cases arose, face-to-face meetings were arranged. Boundary roles (once created) that deal with heterogeneous elements must contain a minimal degree of routinization to maximize flexibility in dealing with special cases.

The degree to which boundary roles are routinized thus is a function of both the need to adapt to environment contingencies and constraints, and the need to control behavior of potentially deviant members. Routinization can serve as a social control mechanism when the organization does not or cannot assume normative commitment of members to organizational procedures. Similar mechanisms would be the use of uniforms to reinforce organizational identification, or frequently shifting employees between boundary roles and core roles to prevent development of identification with elements in the environment (27, p. 21).

H10: *Boundary role routinization will vary directly with the volume of repetitive work, the predictability of outcomes, the homogeneity and stability of the environment, and the need to control the behavior of organizational members.*

Power in Boundary Roles

Thompson (62) noted that where the environment is heterogeneous and shifting and where contingencies are important to the organization, boundary personnel are expected to exercise discretion and develop expertise, and to the extent they are successful in recognizing contingencies, they may become powerful within the organization. The potential power position of boundary spanners was evident in discussion of their information-processing function.

The information that filters into the organization through boundary positions is often not raw data, but the inferences of boundary role incumbents. This type of information is difficult for anyone removed from the boundary to verify. The process of uncertainty absorption is a case of creation of organizational intelligence; and once created, intelligence tends to be accepted (66).

The organization thus relies upon the expertise and discretion of its boundary role personnel. They have a gatekeeper's power, and may become even more powerful if they make correct inferences and if the information is vital for organizational survival (40). Their power is further enhanced to the degree that the nature of the task assigned the boundary role makes routinization of the role difficult, if not impossible.

Pettigrew (49, p. 190) showed how the "self-interested filtering of information during a decision process by a gatekeeper" enabled a boundary spanner to consolidate and enhance his or her power. To the extent that information access and control is a power resource, boundary spanners are in an excellent structural position to convert this resource into actual power.

Labor negotiators provide an example of a boundary role that is difficult to routinize, thus leading to de facto concentration of power in the role. Even though negotiators may deal with fairly homogeneous groups, the outcomes are not highly predictable and the costs to the organization may be high. Therefore, negotiators require some degree of discretionary power. Their power is enhanced to the extent that the group they are negotiating with is powerful.

Specifically, a number of components of union power that derive from the tactics or activities of the union - involvement in city elections, use of strike threats in bargaining - all are associated with a higher degree of power in the boundary unit in city governments (34, p. 27).

H11: *The power of boundary role incumbents will vary inversely with boundary role rou-*

tinization, and directly with their own expertise in accomplishing role requirements and with the costliness and unpredictability of interorganizational transactions.

Organizational dependence on boundary role personnel raises the issue of their commitment to and integration into the organization. The least costly monitoring mechanism is for the organization to rely on the professional identification and ethics of the boundary personnel. More obtrusive strategies include attempts to indoctrinate boundary personnel in organizational policies, norms, and goals, prior to their engaging in interorganizational contacts. Rotation of members among boundary roles is another active strategy, although it has costs in terms of disrupting local adaptations that have been made by boundary spanners. An organization might grant powerful boundary personnel higher positions within the organizational structure to reinforce commitment, although such positions may be a result of the power these members develop through successful interaction on behalf of the organization, e.g. the common practice of picking top management out of the sales division of an organization.

Many studies emphasize the stress and conflict felt by personnel in boundary roles (33, 46), but overlook the positive potential inherent in their role accumulation prospects. Sieber (56) has recently argued that multiple relationships with diverse role partners provide numerous sources of gratification, rather than strain, to individuals such as boundary personnel. He notes that role rights and privileges may accumulate more rapidly than duties, that overall status security may be enhanced by means of buffer roles, that multiple roles can serve as resources for status enhancement and role performance, and finally that multiple roles may enrich one's personality and enhance one's self-conception. While this article is not addressed to the issue of costs and benefits to individuals who occupy boundary roles, the positive side of boundary spanning activities

should be seen as a counter to the negative image currently portrayed in the literature.

Two recent studies in a research and development organization and a large manufacturing company report positive correlations between boundary spanning activity and several dimensions of job satisfaction (34, 35). These studies also found very small or insignificant correlations between role conflict, role ambiguity, and boundary spanning activity. The authors argue that boundary spanning jobs, to the extent they enable role incumbents to reduce uncertainties for others, permit boundary spanners to gain power, improve their bargaining position, and hence increase their job satisfaction and perhaps even gain better jobs.

Implications

The picture of boundary spanning roles portrayed in this article has two implications for the study of formal organizations. First, this view of organization-environment interaction is a decidedly disaggregated one, in contrast to current literature which sees organizations responding as "wholes" to environmental influence (15, 25). We treat boundary spanning roles as the critical link between environmental characteristics and

organizational structure, with the further stipulation that organizations face multiple environments and thus can have a variety of boundary roles of units with different structural characteristics. This implies, for example, that when an investigator studies the impact of interorganizational dependence on organizational structure, the place to begin is with its impact on boundary spanning roles in the immediate vicinity of the dependence relationship, rather than with the structure of the organization as a whole (8, 43).

Second, more empirical studies are needed of how personnel in boundary spanning units or roles carry out their duties, and in particular how such role performance varies under different environmental conditions and over time. This would mean more studies of the type carried out by Mintzberg (44), on the day-to-day behavior of managers, or Mathiesen (41) on the day-to-day behavior of staff members in two Scandinavian prisons. Both studies make extensive use of non-participant observation and detailed first-hand knowledge of the actual, rather than self-reported, behavior of those persons studied. The cumulation of such studies would enable us to understand the process by which boundary spanning roles are generated, elaborated, and used by their incumbents.

REFERENCES

1. Adair, Ross. "The Indian Health Worker," *Human Organization*, Vol. 19 (1960), 59-63.
2. Adams, J. Stacy. "The Structure and Dynamics of Behavior in Organizational Boundary Roles," in M. Dunette (Ed.), *Handbook of Organizational and Industrial Psychology* (Chicago: Rand McNally, 1976), pp. 1175-1199.
3. Aguiler, Francis. *Scanning the Business Environment* (New York: MacMillan, 1967).
4. Aiken, Michael, and Jerald Hage. "Organizational Interdependence and Intra-organizational Structure," *American Sociological Review*, Vol. 33 (1968), 912-930.
5. Aiken, Michael, and Jerald Hage. "Organizational Permeability." Paper presented at the 1972 Meetings of the American Sociological Association.
6. Aldrich, Howard. "Organizational Boundaries and Inter-organizational Conflict," *Human Relations*, Vol. 24 (1971), 279-287.
7. Aldrich, Howard. "An Organization-Environment Perspective on Cooperation and Conflict in the Manpower Training System," in A. Negandhi (Ed.), *Conflict and Power in Complex Organizations* (Kent, Ohio: C.A.R.I., Kent State University, 1972), pp. 11-37.

8. Aldrich, Howard. "Organization Sets, Action Sets, and Networks: Making the Most of Simplicity," in P. Nystrom and W. Starbuck (Eds.), *Handbook of Organizational Design* (Amsterdam: Elsevier, 1977).
9. Aldrich, Howard, and Jeffrey Pfeffer. "Environment of Organizations," in A. Inkeles (Ed.), *Annual Review of Sociology*, Vol. II (Palo Alto, Calif.: Annual Review Inc., 1976), pp. 79-105.
10. Aldrich, Howard, and Albert J. Reiss, Jr. "Police Officers as Boundary Personnel," in H. Hahn (Ed.), *Police in Urban Society* (Beverly Hills, Calif.: Sage Publications, 1971), pp. 193-208.
11. Bennis, Warren. *Beyond Bureaucracy* (New York: McGraw-Hill, 1966).
12. Benson, J. Kenneth. "Models of Structure Selection." Unpublished paper, University of Missouri-Columbia, 1971.
13. Benson, J. Kenneth. "The Interorganizational Network as a Political Economy," *Administrative Science Quarterly*, Vol. 20 (1975), 229-249.
14. Blau, Peter. *The Dynamics of Bureaucracy* (Chicago: University of Chicago Press, 1957).
15. Blau, Peter, and Richard Schoenherr. *The Structure of Organizations* (New York: Basic Books, 1971).
16. Blau, Peter, and W. Richard Scott. *Formal Organizations* (San Francisco: Chandler, 1962).
17. Brown, W. B. "Systems, Boundaries, and Information Flow," *Academy of Management Journal*, Vol. 9 (1966), 318-327.
18. Campbell, Donald. "Variation and Selective Retention in Socio-Cultural Evolution," *General Systems*, Vol. 16 (1969), 69-85.
19. Chandler, Alfred. *Strategy and Structure*, (Cambridge: MIT Press, 1962).
20. Child, John. *The Business Enterprise in Modern Industrial Society*, (London: Collier MacMillan, 1969).
21. Child, John. "Parkinson's Progress: Accounting for the Number of Specialists in Organizations," *Administrative Science Quarterly*, Vol. 18 (1973), 328-348.
22. Child, John. "Participation, Organization and Social Cohesion," *Human Relations*, forthcoming.
23. Cyert, Richard and James March. *A Behavioral Theory of the Firm* (Englewood Cliffs, N.J.: Prentice-Hall, 1963).
24. Dill, William. "The Impact of Environment on Organizational Development," in S. Mailick and E. Van Ness (Eds.), *Concepts and Issues in Administrative Behavior* (Englewood Cliffs, N.J.: Prentice-Hall, 1962).
25. Emery, Fred, and Eric Trist. "The Causal Texture of Organizational Environments," *Human Relations*, Vol. 18 (1965), 21-31.
26. Goffman, Erving. "On the Characteristics of Total Institutions," in D. Cressey (Ed.), *The Prison* (New York: Holt, Rinehart, and Winston, 1961).
27. Guetzkow, Harold. "Relations Among Organizations," in R. Bowers (Ed.), *Studies on Behavior in Organizations* (Athens, Georgia: University of Georgia Press, 1966).
28. Hage, Jerald, and Michael Aiken. *Social Change in Complex Organizations* (New York: Random House, 1970).
29. Harrison, Michael. "The Adjustment of a Social Movement to its Organizational Environment." Unpublished paper, Department of Sociology, SUNY, Stony Brook, 1972.
30. Hasenfeld, Yeheskel. "People Processing Organizations: An Exchange Approach," *American Sociological Review*, Vol. 37 (1972), 256-263.
31. Hickson, David. "A Convergence in Organization Theory," *Administrative Science Quarterly*, Vol. 2 (1966), 224-237.
32. Hirsch, Paul. "Processing Fads and Fashions: An Organizational Set Analysis of Cultural Industry Systems," *American Journal of Sociology* Vol. 77 (1972), 639-659.
33. Katz, Daniel, and Robert Kahn. *The Social Psychology of Organizations* (New York: John Wiley and Sons, 1966).
34. Keller, Robert, and Winford Holland. "Boundary-Spanning Roles in a Research and Development Organization: An Empirical Investigation," *Academy of Management Journal*, Vol. 18 (1975), 388-393.
35. Keller, Robert, Andrew Szilagyi, and Winford Holland. "Boundary-Spanning Activity and Employee Reactions: An Empirical Study," *Human Relations*, forthcoming.
36. Kochan, Thomas. "Determinants of the Power of Boundary Units in an Interorganizational Bargaining Relation," *Administrative Science Quarterly*, Vol. 20 (1975), 434-452.
37. Lawrence, Paul, and Jay Lorsch. *Organization and Environment* (Cambridge, Mass.: Graduate School of Business Administration, Harvard University, 1967).
38. MacAulay, Stewart. "Non-contractual Relations in Business: A Preliminary Study," *American Sociological Review*, Vol. 28 (1963), 55-67.
39. Maniha, John, and Charles Perrow. "The Reluctant Organization and the Aggressive Environment," *Administrative Science Quarterly*, Vol. 10 (1965), 238-257.
40. March, James, and Herbert Simon. *Organizations* (New York: John Wiley and Sons, 1958).
41. Mathieson, Thomas. *Across the Boundaries of Organizations* (Calif: Glendessary Press, 1972).
42. Meier, Richard. "Information Input Overload," in F. Massarik and P. Ratoosh (Eds.), *Mathematical Explorations in Behavioral Sciences* (Homewood, Ill.: Richard Irwin, Inc., 1965).
43. Mindlin, Sergio, and Howard Aldrich. "Interorganizational Dependence: A Review of the Concept and a Re-examination of the Findings of the Aston Group," *Administrative Science Quarterly*, Vol. 20 (1975), 382-392.
44. Mintzberg, Henry. *The Nature of Managerial Work* (New York: Harper and Row, 1973).
45. Nadel, Mark. *The Politics of Consumer Protection* (Indianapolis: Bobbs-Merrill, 1971).

46. Organ, Dennis. "Linking Pins Between Organizations and Environment," *Business Horizons*, Vol. 14 (December 1971), 73-80.
47. Parsons, Talcott. "Suggestions for a Sociological Approach to Theory of Organizations," *Administrative Science Quarterly*, Vol. 1 (1956), 63-69.
48. Perrow, Charles. *Organizational Analysis* (Belmont, Calif.: Wadsworth Publishing Company, 1970).
49. Pettigrew, Andrew. "Information Control as a Power Resource," *Sociology*, Vol. 6 (1972), 187-204.
50. Pfeffer, Jeffrey. "Size and Composition of Corporate Boards of Directors," *Administrative Science Quarterly*, Vol. 17 (1972), 218-228.
51. Pfeffer, Jeffrey, and Huseyin Leblebici. "Executive Recruitment and the Development of Interfirm Organization," *Administrative Science Quarterly*, Vol. 18 (1973), 449-461.
52. Pfeffer, Jeffrey, and Huseyin Leblebici. "The Effect of Competition on Some Dimensions of Organizational Structure," *Social Forces*, Vol. 52 (1973), 268-279.
53. Price, James. "The Impact of Governing Boards on Organizational Effectiveness and Morale," *Administrative Science Quarterly*, Vol. 8 (1963), 361-368.
54. Pruden, H., and R. Reese. "Interorganizational Role-Set Relations and the Performance and Satisfaction of Industrial Salesmen," *Administrative Science Quarterly*, Vol. 17 (1972), 601-609.
55. Reiss, Albert J. *The Police and the Public* (New Haven: Yale University Press, 1971).
56. Sieber, Sam. "Toward a Theory of Role Accumulation," *American Sociological Review*, Vol. 39 (1974), 567-578.
57. Smigel, Erwin. "The Impact of Recruitment on the Large Law Firm," *Human Relations*, Vol. 25 (1960), 56-65.
58. Starbuck, William. "Organizations and Their Environments," in M. Dunnette (Ed.), *Handbook of Organizational and Industrial Psychology* (Chicago: Rand McNally, 1976), pp. 1069-1124.
59. Strauss, George. "Tactics of Lateral Relationship: The Purchasing Agent," *Administrative Science Quarterly*, Vol. 7 (1962), 161-186.
60. Terryberry, Shirley. "The Evolution of the Organizational Environments," *Administrative Science Quarterly*, Vol. 12 (1968), 590-614.
61. Thompson, James. "Organizations and Output Transactions," *American Journal of Sociology*, Vol. 68 (1962), 309-325.
62. Thompson, James. *Organizations in Action* (New York: McGraw-Hill 1967).
63. Thompson, James, and William McEwen. "Organizational Goals and Environment," *American Sociological Review*, Vol. 23 (1958), 23-31.
64. Weber, Max. *The Theory of Social and Economic Organization* (New York: The Free Press, 1947).
65. Whetten, David. *Predicting Organization-Set Dimensions: An Inter-organizational Study of the Effectiveness of Manpower Programs in New York State* (Ph.D. Thesis, Cornell University, 1974).
66. Wilensky, Harold. *Organizational Intelligence* (New York: Basic Books, 1967).
67. Wilson, James Q. *Political Organizations* (New York: Basic Books, 1973).
68. Zald, Mayer. "The Power and Functions of Boards of Directors," *American Journal of Sociology*, Vol. 75 (1969), 97-111.